Разбор статьи

Шаблон с гугл през взял

RETHINKING THE VALUE OF NETWORK PRUNING

Network

Pruning

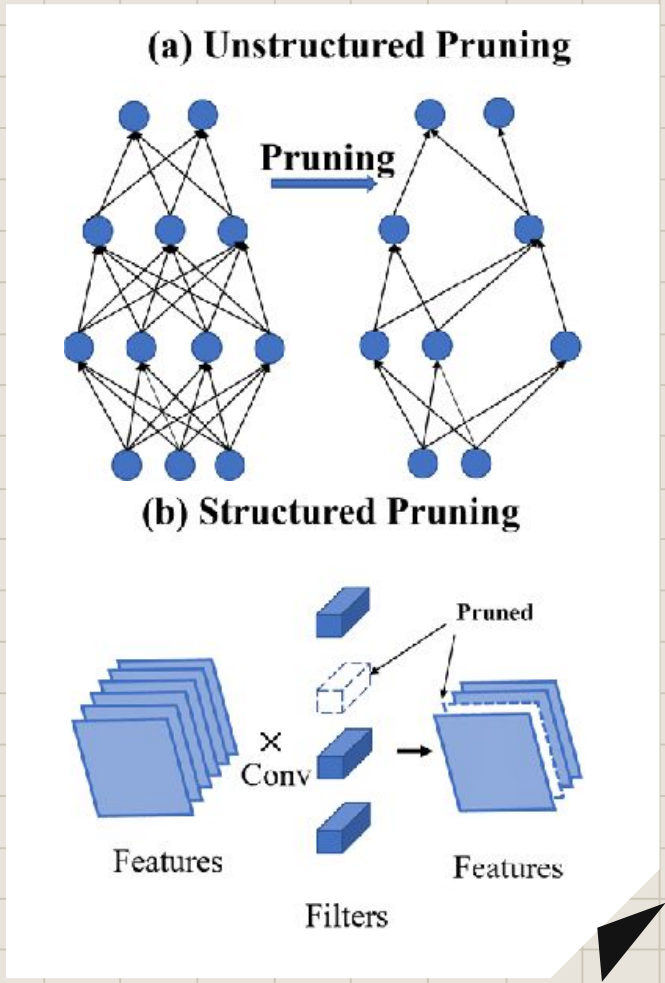
INTRO

Pruning — техника удаления весов/слоёв deep nn для уменьшения затрат на инференс

В работе опровергается несколько популярных гипотез по прунингу:

1. обучение большой, over-parameterized модели не обязательно для эффективной модели
2. выученные “важные” веса большой модели как правило, не дают преимущества для pruned модели
3. для эффективной финальной модели при прунинге важнее полученная архитектура, нежели чем выученные “важные” веса → прунинг может быть полезным в поиске архитектур

В статье также показываются ограничения гипотезы “Lottery Ticket”, в которой говорится, что в deep nn есть выигрышные подсети, равные по качеству оригиналу



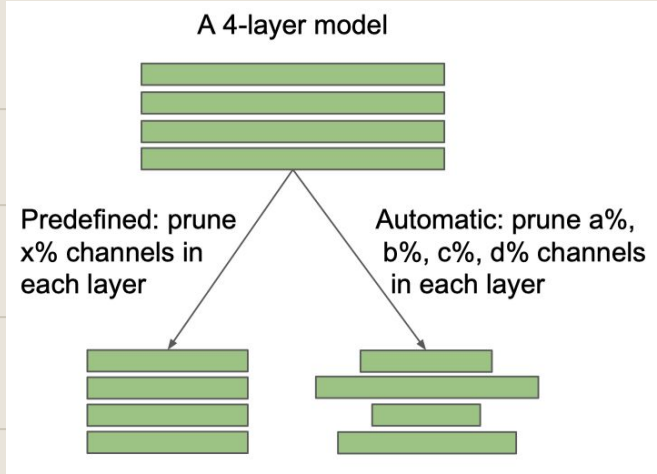
PRUNING

Мифы:

- обучение с over-parameterized модели важно, тк оно дает высокое качество
- pruned архитектура и соответствующие веса важны для эффективной модели

Основной вывод из работы — для структурного прунинга с предзаготовленной архитектурой или autodiscover, обучение со случайной инициализации сравнимого или лучшего чем 3-step пайплайн

Predefined vs Autodiscovered architectures, in channel pruning



3-step пайплайн

Тренировка

большой оверпарамет-ризированной модели

Прунинг

Зануление весов / удаление каналов

Файн-тюнинг

Для восстановлени я качества

METHODOLOGY

Архитектуры:

- VGG
- ResNet
- DenseNet

Датасеты:

- CIFAR-10
- CIFAR-100
- ImageNet

*CIFAR-10 — запускался 5 раз с рандомным сидом и считался mean(acc), std(acc)

Методы прунинга:

- 4x predefined structured
- 2x autodiscover structured
- 1x unstructured

Методы тренировки:

- Scratch-E — equal train epochs
- Scratch-B — equal train flops

*между over-parameterized и pruned моделью

Пайплайн обучения и сравнения:

- Over-parameterized → Pruned
- Pruned → Fine-Tune
- Pruned → Train Scratch-E from random init
- Pruned → Train Scratch-B from random init

EXPERIMENTS – PREDEFINED STRUCTURED PRUNING

L1-norm based Filter Pruning

Dataset	Model	Unpruned	Pruned Model	Fine-tuned	Scratch-E	Scratch-B
CIFAR-10	VGG-16	93.63 (±0.16)	VGG-16-A	93.41 (±0.12)	93.62 (±0.11)	**93.78** (±0.15)
	ResNet-56	93.14 (±0.12)	ResNet-56-A	92.97 (±0.17)	92.96 (±0.26)	**93.09** (±0.14)
			ResNet-56-B	92.67 (±0.14)	92.54 (±0.19)	**93.05** (±0.18)
	ResNet-110	93.14 (±0.24)	ResNet-110-A	93.14 (±0.16)	**93.25** (±0.29)	93.22 (±0.22)
			ResNet-110-B	92.69 (±0.09)	92.89 (±0.43)	**93.60** (±0.25)
ImageNet	ResNet-34	73.31	ResNet-34-A	72.56	72.77	**73.03**
			ResNet-34-B	72.29	72.55	**72.91**

Table 1: Results (accuracy) for L_1-norm based filter pruning (Li et al., 2017). "Pruned Model" is the model pruned from the large model. Configurations of Model and Pruned Model are both from the original paper.

EXPERIMENTS – PREDEFINED STRUCTURED PRUNING

ThiNet

Dataset	Unpruned	Strategy	Pruned Model		
ImageNet	VGG-16		VGG-Conv	VGG-GAP	VGG-Tiny
	71.03	Fine-tuned	−1.23	−3.67	−11.61
	71.51	Scratch-E	−2.75	−4.66	−14.36
		Scratch-B	+**0.21**	−**2.85**	−**11.58**
	ResNet-50		ResNet50-30%	ResNet50-50%	ResNet50-70%
	75.15	Fine-tuned	−6.72	−4.13	−3.10
	76.13	Scratch-E	−5.21	−2.82	−1.71
		Scratch-B	−**4.56**	−**2.23**	−**1.01**

Table 2: Results (accuracy) for ThiNet (Luo et al., 2017). Names such as "VGG-GAP" and "ResNet50-30%" are pruned models whose configurations are defined in Luo et al. (2017). To accommodate the effects of different frameworks between our implementation and the original paper's, we compare relative accuracy drop from the unpruned large model. For example, for the pruned model VGG-Conv, −1.23 is relative to 71.03 on the left, which is the reported accuracy of the unpruned large model VGG-16 in the original paper; −2.75 is relative to 71.51 on the left, which is VGG-16's accuracy in our implementation.

EXPERIMENTS – PREDEFINED STRUCTURED PRUNING

Regression based Feature Reconstruction

Dataset	Unpruned	Strategy	Pruned Model
ImageNet	VGG-16		VGG-16-5x
	71.03	Fine-tuned	−2.67
	71.51	Scratch-E	−3.46
		Scratch-B	**−0.51**
	ResNet-50		ResNet-50-2x
	75.51	Fine-tuned	−3.25
	76.13	Scratch-E	−1.55
		Scratch-B	**−1.07**

Table 3: Results (accuracy) for Regression based Feature Reconstruction (He et al., 2017b). Pruned models such as "VGG-16-5x" are defined in He et al. (2017b). Similar to Table 2, we compare relative accuracy drop from unpruned large models.

EXPERIMENTS – AUTOMATIC STRUCTURED PRUNING

Network Slimming

Dataset	Model	Unpruned	Prune Ratio	Fine-tuned	Scratch-E	Scratch-B
CIFAR-10	VGG-19	93.53 (±0.16)	70%	93.60 (±0.16)	93.30 (±0.11)	**93.81** (±0.14)
	PreResNet-164	95.04 (±0.16)	40%	94.77 (±0.12)	94.70 (±0.11)	**94.90** (±0.04)
			60%	94.23 (±0.21)	94.58 (±0.18)	**94.71** (±0.21)
	DenseNet-40	94.10 (±0.12)	40%	94.00 (±0.20)	93.68 (±0.18)	**94.06** (±0.12)
			60%	**93.87** (±0.13)	93.58 (±0.21)	93.85 (±0.25)
CIFAR-100	VGG-19	72.63 (±0.21)	50%	72.32 (±0.28)	71.94 (±0.17)	**73.08** (±0.22)
	PreResNet-164	76.80 (±0.19)	40%	76.22 (±0.20)	76.36 (±0.32)	**76.68** (±0.35)
			60%	74.17 (±0.33)	75.05 (± 0.08)	**75.73** (±0.29)
	DenseNet-40	73.82 (±0.34)	40%	**73.35** (±0.17)	73.24 (±0.29)	73.19 (±0.26)
			60%	72.46 (±0.22)	72.62 (±0.36)	**72.91** (±0.34)
ImageNet	VGG-11	70.84	50%	68.62	70.00	**71.18**

Table 4: Results (accuracy) for Network Slimming (Liu et al., 2017). "Prune ratio" stands for total percentage of channels that are pruned in the whole network. The same ratios for each model are used as the original paper.

EXPERIMENTS – AUTOMATIC STRUCTURED PRUNING

Sparse Structure Selection

Dataset	Model	Unpruned	Pruned Model	Pruned	Scratch-E	Scratch-B
ImageNet	ResNet-50	76.12	ResNet-41	75.44	75.61	**76.17**
			ResNet-32	74.18	73.77	**74.67**
			ResNet-26	71.82	72.55	**73.41**

Table 5: Results (accuracy) for residual block pruning using Sparse Structure Selection (Huang & Wang, 2018). In the original paper no fine-tuning is required so there is a "Pruned" column instead of "Fine-tuned" as before.

EXPERIMENTS – UNSTRUCTURED MAGNITUDE-BASED PRUNING

Dataset	Model	Unpruned	Prune Ratio	Fine-tuned	Scratch-E	Scratch-B
CIFAR-10	VGG-19	93.50 (±0.11)	30%	93.51 (±0.05)	**93.71** (±0.09)	93.31 (±0.26)
			80%	93.52 (±0.10)	**93.71** (±0.08)	93.64 (±0.09)
			95%	93.34 (±0.13)	93.21 (±0.17)	**93.63** (±0.18)
	PreResNet-110	95.04 (±0.15)	30%	95.06 (±0.05)	94.84 (±0.07)	**95.11** (±0.09)
			80%	**94.55** (±0.11)	93.76 (±0.10)	94.52 (±0.13)
			95%	**92.35** (±0.20)	91.23 (±0.11)	91.55 (±0.34)
	DenseNet-BC-100	95.24 (±0.17)	30%	95.21 (±0.17)	95.22 (±0.18)	**95.23** (±0.14)
			80%	95.04 (±0.15)	94.42 (±0.12)	**95.12** (±0.04)
			95%	**94.19** (±0.15)	92.91 (±0.22)	93.44 (±0.19)
CIFAR-100	VGG-19	71.70 (±0.31)	30%	71.96 (±0.36)	72.81 (±0.31)	**73.30** (±0.25)
			50%	71.85 (±0.30)	73.12 (±0.36)	**73.77** (±0.23)
			95%	70.22 (±0.38)	70.88 (±0.35)	**72.08** (±0.15)
	PreResNet-110	76.96 (±0.34)	30%	76.88 (±0.31)	76.36 (±0.26)	**76.96** (±0.31)
			50%	**76.60** (±0.36)	75.45 (±0.23)	76.42 (±0.39)
			95%	68.55 (±0.51)	68.13 (±0.64)	**68.99** (±0.32)
	DenseNet-BC-100	77.59 (±0.19)	30%	77.23 (±0.05)	77.58 (±0.25)	**77.97** (±0.31)
			50%	77.41 (±0.14)	77.65 (±0.09)	**77.80** (±0.23)
			95%	**73.67** (±0.03)	71.47 (±0.46)	72.57 (±0.37)
ImageNet	VGG-16	73.37	30%	73.68	72.75	**74.02**
			60%	**73.63**	71.50	73.42
	ResNet-50	76.15	30%	**76.06**	74.77	75.70
			60%	**76.09**	73.69	74.91

Table 6: Results (accuracy) for unstructured pruning (Han et al., 2015). "Prune Ratio" denotes the percentage of parameters pruned in the set of all convolutional weights.

NETWORK PRUNING AS ARCHITECTURE SEARCH

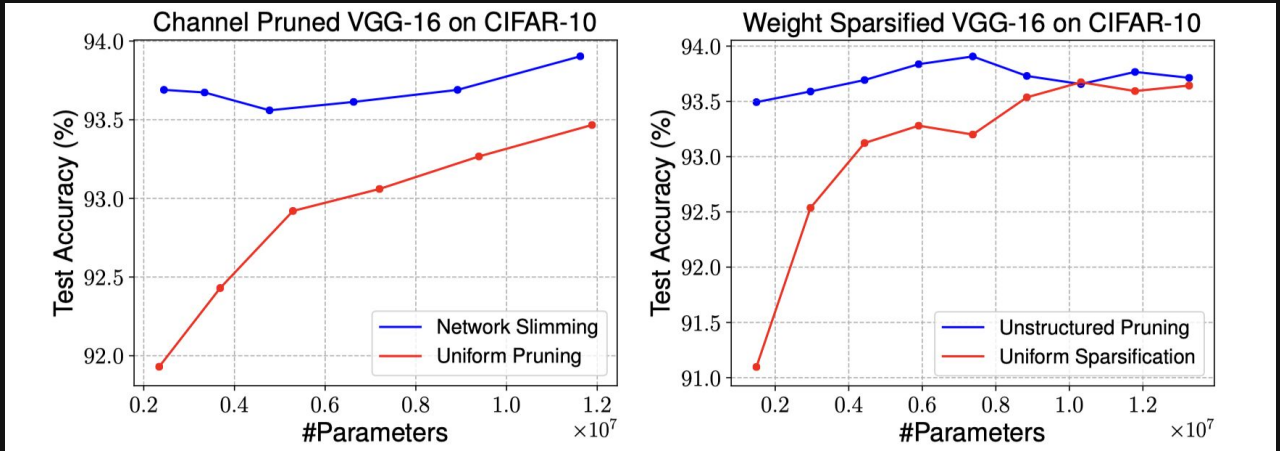


Figure 3: Pruned architectures obtained by different approaches, all *trained from scratch*, averaged over 5 runs. Architectures obtained by automatic pruning methods (*Left:* Network Slimming (Liu et al., 2017), *Right:* Unstructured pruning (Han et al., 2015)) have better parameter efficiency than uniformly pruning channels or sparsifying weights in the whole network.

Для structured pruning важнее не унаследованные веса, а полученная архитектура

Automatic pruning можно рассматривать как поиск эффективной архитектуры

Если архитектура уже известна, её можно обучить с нуля

Для VGG16 pruning сильно повышает эффективность модели

NETWORK PRUNING AS ARCHITECTURE SEARCH

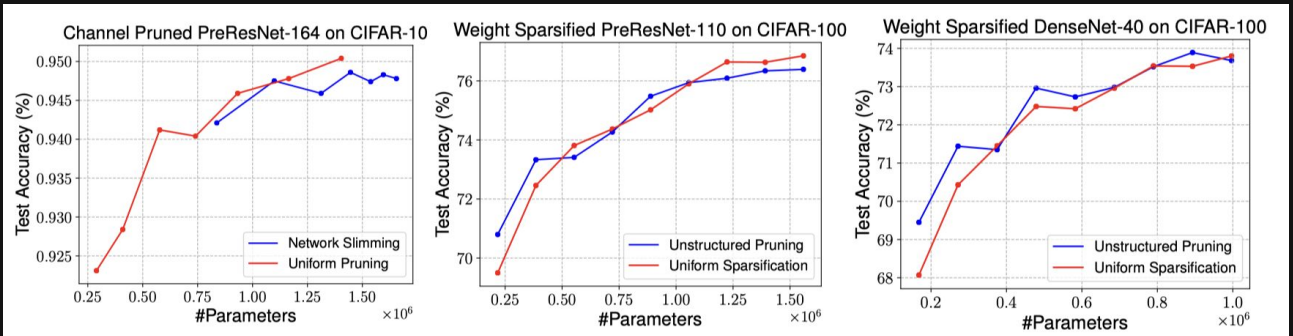
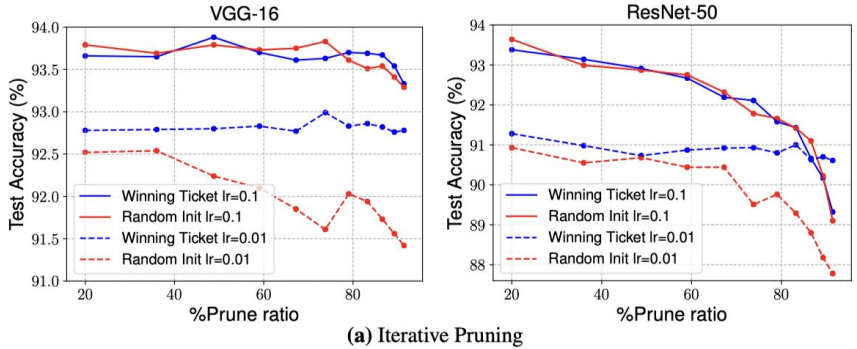


Figure 5: Pruned architectures obtained by different approaches, *all trained from scratch*, averaged over 5 runs. *Left:* Results for PreResNet-164 pruned on CIFAR-10 by Network Slimming (Liu et al., 2017). *Middle* and *Right*: Results for PreResNet-110 and DenseNet-40 pruned on CIFAR-100 by unstructured pruning (Han et al., 2015).

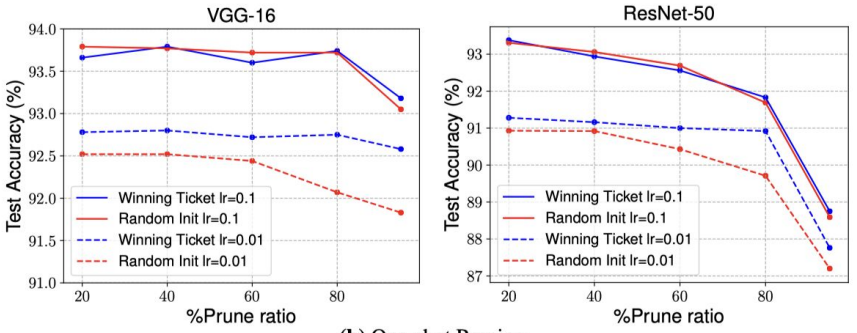
На VGG pruning-архитектуры заметно лучше uniform pruning, а на ResNet / DenseNet преимущество часто малое или почти отсутствует

Поэтому pruning как architecture search полезен не всегда одинаково

LOTTERY TICKET HYPOTHESIS (FRANKLE & CARBIN, 2019)



(a) Iterative Pruning



(b) One-shot Pruning

Figure 7: Comparisons with the Lottery Ticket Hypothesis (Frankle & Carbin, 2019) for iterative/one-shot unstructured pruning (Han et al., 2015) with two initial learning rates 0.1 and 0.01, on CIFAR-10 dataset. Each point is averaged over 5 runs. Using the winning ticket as initialization only brings improvement when the learning rate is small (0.01), however such small learning rate leads to a lower accuracy than the widely used large learning rate (0.1).

На графиках сравнение обучения подсетей с исходной и случайной инициализацией весов

При маленьком learning rate в 0.01 исходная инициализация даёт преимущество

При большом learning rate 0.1 преимущество исчезает, а итоговое качество выше

Для structured pruning исходная инициализация не даёт преимущества независимо от learning rate

THANK YOU!